Elementary Analysis

Chapter 3, Section 17

Continuous Functions
Exercises 17.1–17.16, with every subpart

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 3. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Contents

17 Continuous functions **3**

17.1 Exercise 17.1: Domains and values of two compositions 3
17.2 Exercise 17.2: Compositions can remove or preserve a jump 4
17.3 Exercise 17.3: Continuity from algebra and composition 5
17.4 Exercise 17.4: Continuity of the square root including its endpoint 6
17.5 Exercise 17.5: Powers and polynomials are continuous 6
17.6 Exercise 17.6: Continuity of rational functions on their domains 7
17.7 Exercise 17.7: Linear maps, absolute value, and scalar multiples 7
17.8 Exercise 17.8: Formulas and continuity for a pointwise minimum 8
17.9 Exercise 17.9: Four explicit epsilon–delta proofs 9
17.10 Exercise 17.10: Three explicit sequences witnessing discontinuity 10
17.11 Exercise 17.11: Testing continuity using only monotone sequences 10
17.12 Exercise 17.12: A continuous function is determined on a dense set 11
17.13 Exercise 17.13: Rational and irrational inputs with different values 11

17.14 Exercise 17.14: A denominator-sensitive function on the rationals 12
17.15 Exercise 17.15: A continuity test that omits the point itself 13
17.16 Exercise 17.16: Jumps in the postage-stamp function 14

17 Continuous functions

Continuity is always relative to a function's domain. At a point x_0 in that domain, Definition 17.1 requires $f(x_n) \to f(x_0)$ for every domain sequence $x_n \to x_0$. The equivalent epsilon–delta condition of Theorem 17.2 tests only points x in the domain as well. This qualification matters for square roots, quotients, and isolated domain points.

Two simple starting functions are continuous on $\mathbb{R}$: the identity $I(x) = x$ and every constant function $C(x) = c$. Indeed, for $x_n \to x_0$, one has $I(x_n) = x_n \to x_0 = I(x_0)$ and $C(x_n) = c \to c = C(x_0)$. Theorems 17.3–17.5 then supply scalar multiples, sums, products, quotients with nonzero denominators, and compositions. Each application below checks the domain or sign condition it needs.

17.1 Exercise 17.1: Domains and values of two compositions

What we must show. For $f(x) = \sqrt{4-x}$ on $(-\infty, 4]$ and $g(x) = x^2$ on $\mathbb{R}$, answer all four domain and composition questions.

Plan. A sum or product needs both original values. A composition also needs the inner value to belong to the outer function's domain.

(a) Find the domains before evaluating anything.

For $f + g$ and fg, intersect the two domains:

$$\operatorname{dom}(f+g) = \operatorname{dom}(fg) = (-\infty, 4] \cap \mathbb{R} = (-\infty, 4].$$

Their formulas there are $\sqrt{4-x} + x^2$ and $x^2\sqrt{4-x}$, respectively.

For $f \circ g$, the inner function g is defined for every real x, but its value must satisfy $g(x) \le 4$. This means $x^2 \le 4$, or equivalently $-2 \le x \le 2$. Hence

$$(f \circ g)(x) = \sqrt{4-x^2}, \qquad \operatorname{dom}(f \circ g) = [-2, 2].$$

For $g \circ f$, first $x \le 4$ is required to define $f(x)$. The value $f(x)$ is then real, and g accepts every real input. There is no additional restriction. Squaring the nonnegative square root gives

$$(g \circ f)(x) = (\sqrt{4-x})^2 = 4 - x, \qquad \operatorname{dom}(g \circ f) = (-\infty, 4].$$

(b) Substitute the requested inputs into the appropriate formulas.

All three inputs $0, 1, 2$ lie in both composition domains. Their values, including the intermediate arithmetic, are

x	$(f \circ g)(x) = \sqrt{4-x^2}$	$(g \circ f)(x) = 4 - x$
0	$\sqrt{4-0} = 2$	$4 - 0 = 4$
1	$\sqrt{4-1} = \sqrt{3}$	$4 - 1 = 3$
2	$\sqrt{4-4} = 0$	$4 - 2 = 2$

Thus the six values in the order requested are $2, 4, \sqrt{3}, 3, 0, 2$.

(c) Check equality as functions, not only equality of formulas.

The functions have different domains. Moreover, even on their common domain they differ: at 0 the first has value 2 and the second value 4. Either observation disproves equality.

(d) Test whether the inner value is admissible at 3.

Since $g(3) = 9 > 4$, the expression $f(g(3)) = f(9)$ is not defined as a real value. In contrast, $3 \leq 4$, so $f(3) = \sqrt{1} = 1$ is defined and $g(f(3)) = g(1) = 1$ is defined.

Conclusion. The sum and product domains are $(-\infty, 4]$; the two composition domains are $[-2, 2]$ and $(-\infty, 4]$. The compositions are unequal. At 3, only $g \circ f$ is meaningful.

17.2 Exercise 17.2: Compositions can remove or preserve a jump

What we must show. Let $f(x) = 0$ for $x < 0$, $f(x) = 4$ for $x \geq 0$, and $g(x) = x^2$ for every real x. Determine the four combinations in part (a), and decide which of the six functions in part (b) are continuous.

(a) Work separately on the two sides of zero.

Both original domains are $\mathbb{R}$. Sums and products are therefore defined on $\mathbb{R}$, with formulas

$$(f+g)(x) = \begin{cases} x^2, & x < 0, \\ x^2 + 4, & x \geq 0, \end{cases} \qquad (fg)(x) = \begin{cases} 0, & x < 0, \\ 4x^2, & x \geq 0. \end{cases}$$

For every real x, $g(x) = x^2 \geq 0$. Thus $f(g(x)) = 4$, including at $x = 0$, because $f(0) = 4$. Hence

$$(f \circ g)(x) = 4 \qquad (x \in \mathbb{R}).$$

For the opposite order, square the two possible values of f:

$$(g \circ f)(x) = \begin{cases} 0^2 = 0, & x < 0, \\ 4^2 = 16, & x \geq 0. \end{cases}$$

The composition domains are $\mathbb{R}$ as well, since all inner values belong to the outer domains.

(b) First check points other than zero.

At any $x_0 \neq 0$, a radius $|x_0|/2$ keeps x on the same side of zero as x_0. Each displayed function is then given near x_0 by one constant or polynomial formula. Such formulas are continuous by the identity, constant, sum, and product rules. Thus zero is the only possible discontinuity of any function in this exercise.

(b) Identify the three jumps with one sequence.

Take $x_n = -1/n$. Then $x_n \to 0$, but

$$f(x_n) = 0 \to 0 \neq 4 = f(0),$$

$$(f+g)(x_n) = 1/n^2 \to 0 \neq 4 = (f+g)(0),$$

and

$$(g \circ f)(x_n) = 0 \to 0 \neq 16 = (g \circ f)(0).$$

Definition 17.1 therefore makes all three functions discontinuous at zero.

(b) Check the functions that remain continuous at zero.

The function $g(x) = x^2$ is a product of two identity functions, so it is continuous everywhere. The function $f \circ g$ is constant, so it is continuous everywhere too.

For fg, its value at zero is 0, and for every real x, $|(fg)(x)| \leq 4x^2$. Given $\varepsilon > 0$, choose $\delta = \sqrt{\varepsilon}/2 > 0$. If $|x| < \delta$, then

$$|(fg)(x) - (fg)(0)| \leq 4x^2 < 4\delta^2 = \varepsilon.$$

Hence fg is continuous at zero and therefore on all of $\mathbb{R}$.

Conclusion. The globally continuous functions are g, fg, and $f \circ g$. The functions f, $f + g$, and $g \circ f$ are continuous away from zero but discontinuous at zero.

17.3 Exercise 17.3: Continuity from algebra and composition

What we must show. Prove continuity of all eight requested functions using the given familiar continuous functions and the section's rules.

Plan. Decompose each expression into known continuous maps, then check that every outer function receives an allowed input.

(a) The function $\log(1 + \cos^4 x)$.

The given continuity of cosine and repeated products show that $\cos^4 x$ is continuous on $\mathbb{R}$. Addition of the constant 1 preserves continuity. Furthermore, $1 + \cos^4 x \geq 1 > 0$ for every real x. The logarithm is continuous on $(0, \infty)$, which contains all these inner values. The composition theorem therefore proves continuity on the natural domain $\mathbb{R}$.

(b) The function $[\sin^2 x + \cos^6 x]^\pi$.

Products and a sum show that $u(x) = \sin^2 x + \cos^6 x$ is continuous on $\mathbb{R}$. Both summands are nonnegative. They cannot both vanish: if $\sin x = 0$, the identity $\sin^2 x + \cos^2 x = 1$ gives $\cos^2 x = 1$, so $\cos^6 x = 1$. If $\sin x \neq 0$, the first summand is positive instead. Thus $u(x) > 0$ for every x. The given power function $t \mapsto t^\pi$ is continuous for $t > 0$, so its composition with u is continuous on $\mathbb{R}$.

(c) The function 2^{x^2}.

The map $x \mapsto x^2 = I(x)I(x)$ is continuous on $\mathbb{R}$. The given map $t \mapsto 2^t$ is continuous for every real t. Its composition with x^2 is therefore continuous on $\mathbb{R}$.

(d) The function 8^x.

Use $8^x = 2^{3x}$. The inner map $x \mapsto 3x$ is continuous as a scalar multiple of the identity, and the outer map $t \mapsto 2^t$ is continuous on $\mathbb{R}$. Composition proves the claim on $\mathbb{R}$ without needing a new rule for the base 8.

(e) The tangent function on its stated domain.

Write $\tan x = \sin x / \cos x$. Sine and cosine are continuous on $\mathbb{R}$, and cosine is nonzero exactly away from $x = \pi/2 + k\pi$, $k \in \mathbb{Z}$. The quotient theorem proves continuity at every point of that natural domain.

(f) The function $x \sin(1/x)$ for $x \neq 0$.

The constant function 1 and identity are continuous; the quotient $r(x) = 1/x$ is therefore continuous on $D = \mathbb{R} \setminus \{0\}$. Since sine is continuous on all of $\mathbb{R}$, $\sin(r(x))$ is continuous on D by composition. Multiplying by the continuous identity gives continuity of $x \sin(1/x)$ on D.

(g) The function $x^2 \sin(1/x)$ for $x \neq 0$.

The sine factor is continuous on D by part (f)'s construction. The factor x^2 is continuous on $\mathbb{R}$, hence on D. Their product is continuous at every point of D.

(h) The function $x^{-1} \sin(1/x^2)$ for $x \neq 0$.

On D, $x^2 \neq 0$. The quotient $1/x^2$ is continuous there, and its composition with sine is continuous there. The other factor $1/x$ is continuous on D too. Their product is the requested continuous function.

Conclusion. Every function is continuous on its natural stated domain: $\mathbb{R}$ in (a)–(d), the nonzero-cosine domain in (e), and $\mathbb{R} \setminus \{0\}$ in (f)–(h). No value at zero is asserted in the last three parts.

17.4 Exercise 17.4: Continuity of the square root including its endpoint

What we must show. Prove that $f(x) = \sqrt{x}$ is continuous on $[0, \infty)$.

Plan. An estimate valid even at zero avoids dividing by a possibly small sum of square roots.

Prove the estimate for any two domain points.

If $x \geq y \geq 0$, then $\sqrt{x} - \sqrt{y} \geq 0$ and $\sqrt{x} + \sqrt{y} \geq \sqrt{x} - \sqrt{y}$. Multiplication by $\sqrt{x} - \sqrt{y}$ preserves this order, giving

$$x - y = (\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y}) \geq (\sqrt{x} - \sqrt{y})^2.$$

Taking nonnegative square roots yields $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$. If $y > x$, interchange the two variables; the same absolute-value inequality results. Thus it holds for every $x, y \geq 0$.

Choose delta at an arbitrary point of the domain.

Fix $x_0 \geq 0$ and $\varepsilon > 0$, and put $\delta = \varepsilon^2 > 0$. For a domain point $x \geq 0$ with $|x - x_0| < \delta$, the estimate gives

$$|\sqrt{x} - \sqrt{x_0}| \leq \sqrt{|x - x_0|} < \sqrt{\delta} = \varepsilon.$$

This is Theorem 17.2. It includes $x_0 = 0$, because the condition tests only $x \geq 0$ and never requires a negative input.

Conclusion. The square-root function is continuous at every point of its domain, including zero.

17.5 Exercise 17.5: Powers and polynomials are continuous

What we must show. Prove part (a) for every positive integer power, then deduce part (b) for every polynomial.

(a) Build powers by induction using the product rule.

For $m = 1$, $x \mapsto x^m = x$ is the identity, already shown continuous. Suppose $x \mapsto x^m$ is continuous on $\mathbb{R}$. Then $x^{m+1} = x^m x$ is the product of that continuous function and the identity. Theorem 17.4(ii) makes it continuous on $\mathbb{R}$. Mathematical induction proves continuity for all $m \in \mathbb{N}$.

(b) Assemble a polynomial from finitely many continuous terms.

Let

$$p(x) = a_0 + a_1 x + \cdots + a_n x^n, \qquad a_0, \ldots, a_n \in \mathbb{R}.$$

The term a_0 is constant and continuous. For $j \geq 1$, part (a) makes x^j continuous, and Theorem 17.3 makes $a_j x^j$ continuous, including when $a_j = 0$. Theorem 17.4(i) preserves continuity under addition of two terms. Applying it finitely many times proves continuity of the displayed sum at every real point. A constant polynomial is already covered even if there are no positive-power terms.

Conclusion. Every positive integer power and every real polynomial is continuous on $\mathbb{R}$.

17.6 Exercise 17.6: Continuity of rational functions on their domains

What we must show. Let p, q be polynomial functions and define

$$D = \{x \in \mathbb{R} : q(x) \neq 0\}, \qquad f(x) = p(x)/q(x) \quad (x \in D).$$

Prove f is continuous.

Check the hypotheses at an arbitrary allowed point.

Fix $x_0 \in D$. Exercise 17.5 makes both p and q continuous at x_0, and membership in D gives $q(x_0) \neq 0$. The quotient theorem, Theorem 17.4(iii), therefore applies and gives continuity of p/q at x_0.

For a direct reading of that theorem, take any $x_n \in D$ with $x_n \to x_0$. Continuity gives $p(x_n) \to p(x_0)$ and $q(x_n) \to q(x_0) \neq 0$. The quotient limit law gives

$$f(x_n) = \frac{p(x_n)}{q(x_n)} \longrightarrow \frac{p(x_0)}{q(x_0)} = f(x_0).$$

Every denominator in this display is nonzero by the domain conditions. Since $x_0 \in D$ was arbitrary, the function is continuous on D.

Conclusion. A rational function is continuous wherever its denominator does not vanish. Excluded zeros of the denominator are not points at which continuity of this domain function is tested.

17.7 Exercise 17.7: Linear maps, absolute value, and scalar multiples

What we must show. Prove (a) continuity of $x \mapsto kx$, (b) continuity of absolute value, and (c) Theorem 17.3 by composition.

(a) A linear map is a polynomial.

For a fixed real k, the function $G(t) = kt$ is a polynomial with constant coefficient zero and linear coefficient k. Exercise 17.5 makes it continuous on $\mathbb{R}$. This includes $k = 0$, in which case it is constant.

(b) Bound the error in absolute value by the input error.

The triangle inequality gives $|x| \leq |x - x_0| + |x_0|$, so $|x| - |x_0| \leq |x - x_0|$. Interchanging x and x_0 gives $|x_0| - |x| \leq |x - x_0|$. Together these two inequalities are

$$\big||x| - |x_0|\big| \leq |x - x_0|\,.$$

Given $\varepsilon > 0$, choose $\delta = \varepsilon$. Then $|x - x_0| < \delta$ forces $||x| - |x_0|| < \varepsilon$. Thus $H(t) = |t|$ is continuous at every real point.

(c) Compose with the given function.

Suppose f is continuous at a point x_0 of its domain. The outer maps G and H are continuous at the real number $f(x_0)$ by parts (a) and (b). Theorem 17.5 gives continuity at x_0 of

$$(G \circ f)(x) = kf(x), \qquad (H \circ f)(x) = |f(x)|\,.$$

The outer maps have domain $\mathbb{R}$, so both compositions retain the whole domain of f. These are precisely the two functions in Theorem 17.3.

Conclusion. Linear maps and absolute value are continuous on $\mathbb{R}$; composition with them gives a second proof of Theorem 17.3.

17.8 Exercise 17.8: Formulas and continuity for a pointwise minimum

What we must show. Verify both pointwise identities and use one to prove the continuity assertion in part (c).

(a) Verify the absolute-value formula by cases.

For arbitrary real numbers a, b, suppose first that $a \le b$. Then $|a - b| = b - a$, and

$$\frac{a + b - |a - b|}{2} = \frac{a + b - (b - a)}{2} = a = \min\{a, b\}.$$

If $a > b$, then $|a - b| = a - b$, and the same expression is $(a + b - (a - b))/2 = b = \min\{a, b\}$. Substituting $a = f(x)$ and $b = g(x)$ wherever both are defined gives

$$\min(f, g) = \tfrac{1}{2}(f + g) - \tfrac{1}{2}\,|f - g|\,.$$

(b) Verify the negated-maximum formula.

If $a \le b$, negation reverses order, so $-a \ge -b$. Thus $\max\{-a, -b\} = -a$ and its negative is a. If $a > b$, one has $-a < -b$, so the negative of the maximum is b. In both cases, $\min\{a, b\} = -\max\{-a, -b\}$. Pointwise substitution therefore gives $\min(f, g) = -\max(-f, -g)$.

(c) Apply the continuity rules to part (a).

Suppose f and g are continuous at x_0 and both are defined there. Their sum and difference are continuous at x_0 by the sum and scalar rules. The absolute-value rule makes $|f - g|$ continuous there. Multiplying the two terms of part (a) by $1/2$ and $-1/2$, then adding, preserves continuity. The resulting function is $\min(f, g)$. Its domain is the common domain of f and g, so the point x_0 is an allowed point throughout the argument.

Conclusion. Both identities hold, and the pointwise minimum of two functions continuous at x_0 is continuous there.

17.9 Exercise 17.9: Four explicit epsilon–delta proofs

What we must show. Verify the epsilon–delta condition directly in each part, with a stated positive choice of δ.

(a) The square function at $x_0 = 2$.

Fix $\varepsilon > 0$. Factor the error:

$$\left|x^2 - 4\right| = |x - 2|\,|x + 2|\,.$$

To bound the second factor, first require $|x - 2| < 1$. Then $|x + 2| \le |x - 2| + 4 < 5$. Choose

$$\delta = \min\{1, \varepsilon/5\} > 0.$$

If $|x - 2| < \delta$, both preliminary restrictions hold, and $|x^2 - 4| \le 5\,|x - 2| < 5\delta \le \varepsilon$. This proves continuity at 2.

(b) The square-root function at zero.

The domain restriction is $x \ge 0$. Given $\varepsilon > 0$, put $\delta = \varepsilon^2 > 0$. If $x \ge 0$ and $|x - 0| < \delta$, then $0 \le x < \varepsilon^2$ and

$$\left|\sqrt{x} - \sqrt{0}\right| = \sqrt{x} < \varepsilon.$$

The proof tests only domain points, as required at this endpoint.

(c) The function $x\sin(1/x)$ extended by zero.

Let $f(0) = 0$ and $f(x) = x\sin(1/x)$ for $x \ne 0$. For nonzero x, the bound $|\sin(1/x)| \le 1$ gives $|f(x)| \le |x|$. It also holds at $x = 0$, where both sides are zero. Given $\varepsilon > 0$, choose $\delta = \varepsilon$. If $|x| < \delta$, then

$$|f(x) - f(0)| = |f(x)| \le |x| < \varepsilon.$$

Hence f is continuous at zero despite the oscillating sine factor.

(d) The cube function at an arbitrary $x_0 \in \mathbb{R}$.

Fix $\varepsilon > 0$ and set $M = |x_0| + 1 > 0$. If $|x - x_0| < 1$, then $|x| \le |x - x_0| + |x_0| < M$ and $|x_0| < M$. Thus

$$\left|x^2 + xx_0 + x_0^2\right| \le |x|^2 + |x|\,|x_0| + |x_0|^2 \le 3M^2.$$

Using the hinted factorization, choose

$$\delta = \min\left\{1, \frac{\varepsilon}{3M^2}\right\} > 0.$$

For $|x - x_0| < \delta$, the local bound applies, and

$$\left|x^3 - x_0^3\right| = |x - x_0|\left|x^2 + xx_0 + x_0^2\right| \le 3M^2\,|x - x_0| < 3M^2\delta \le \varepsilon.$$

Conclusion. The stated delta choices verify continuity in all four parts directly, with all local bounds imposed before use.

17.10 Exercise 17.10: Three explicit sequences witnessing discontinuity

What we must show. Show discontinuity at zero in parts (a)–(c) by finding domain sequences approaching zero whose function values do not approach the assigned value zero.

(a) The one-sided step function.

Here $f(x) = 1$ for $x > 0$ and $f(x) = 0$ for $x \leq 0$. Take $x_n = 1/n$. Then $x_n > 0$ and $x_n \to 0$, but $f(x_n) = 1$ for all n. The image sequence tends to 1, whereas $f(0) = 0$. This violates Definition 17.1.

(b) The oscillating sine function.

Here $g(x) = \sin(1/x)$ for $x \neq 0$ and $g(0) = 0$. Put

$$x_n = \frac{1}{2\pi n + \pi/2}.$$

The denominator is positive and tends to $+\infty$, so $x_n > 0$ and $x_n \to 0$. At these inputs,

$$g(x_n) = \sin(2\pi n + \pi/2) = 1.$$

Thus $g(x_n)$ tends to 1, not to $g(0) = 0$.

(c) The signum function.

For $\operatorname{sgn}(x) = 1$ at positive inputs, -1 at negative inputs, and 0 at zero, again take $x_n = 1/n$. Then $x_n \to 0$ while $\operatorname{sgn}(x_n) = 1$ for all n, contrary to the image limit required by continuity.

Conclusion. Each function is discontinuous at zero. In every case the displayed sequence stays in the domain and approaches the point, while its image remains at distance 1 from the value there.

17.11 Exercise 17.11: Testing continuity using only monotone sequences

What we must show. Prove that f is continuous at a domain point x_0 if and only if every monotone domain sequence $x_n \to x_0$ satisfies $f(x_n) \to f(x_0)$.

Plan. For the reverse implication, a failure of continuity produces a bad sequence; Theorem 11.4 supplies a monotone subsequence of that sequence without repairing its bad images.

Forward implication: restrict the sequences already tested.

If f is continuous at x_0, Definition 17.1 gives the image limit for every domain sequence approaching x_0. In particular it gives it for every monotone such sequence.

Reverse implication: express failure of continuity precisely.

Assume the stated condition for monotone sequences, and suppose for contradiction that f is not continuous at x_0. The negation of Theorem 17.2 supplies one fixed $\varepsilon_0 > 0$ such that, for every $\delta > 0$, there is u in the domain with

$$|u - x_0| < \delta, \qquad |f(u) - f(x_0)| \geq \varepsilon_0.$$

For each $n \in \mathbb{N}$, use $\delta = 1/n$ and select such a point u_n. Then $|u_n - x_0| < 1/n$ proves $u_n \to x_0$, but every image has error at least ε_0.

Select monotone inputs while retaining their bad images.

Theorem 11.4 gives a monotone subsequence (u_{n_k}). It still lies in the domain, and Theorem 11.3 gives $u_{n_k} \to x_0$. The assumed monotone-sequence criterion would therefore give $f(u_{n_k}) \to f(x_0)$. That is impossible, since

$$|f(u_{n_k}) - f(x_0)| \geq \varepsilon_0 \qquad \text{for every } k.$$

For example, no image can satisfy the eventual error bound $\varepsilon_0/2$. The contradiction proves continuity.

Conclusion. Monotone test sequences suffice. The argument selects a monotone subsequence; it does not assume the original sequence of bad inputs is monotone.

17.12 Exercise 17.12: A continuous function is determined on a dense set

What we must show. Prove (a) that zero values at all rational points of (a, b) force zero values everywhere, and (b) that equality of two continuous functions at those rational points forces equality throughout the interval.

(a) Approximate an arbitrary interval point by rational points inside it.

Fix $x \in (a, b)$. For each $n \in \mathbb{N}$, put

$$\rho_n = \min\{(x - a)/2, (b - x)/2, 1/n\} > 0.$$

The open interval $(x - \rho_n, x + \rho_n)$ lies in (a, b) and has positive length. Rational density supplies $r_n \in \mathbb{Q}$ in that interval. In particular, $|r_n - x| < \rho_n \leq 1/n$, so $r_n \to x$. Since f is continuous at x and $f(r_n) = 0$ for every n,

$$f(x) = \lim_{n\to\infty} f(r_n) = \lim_{n\to\infty} 0 = 0.$$

The point x was arbitrary, so f vanishes everywhere on (a, b).

(b) Reduce equality to the zero-valued case.

The difference $h = f - g$ is continuous on (a, b) by the sum and scalar rules. At every rational point r in that interval, the hypothesis gives $h(r) = f(r) - g(r) = 0$. Part (a), applied to h, yields $h(x) = 0$ for every $x \in (a, b)$. Rearranging $f(x) - g(x) = 0$ gives $f(x) = g(x)$ there.

Conclusion. Both assertions follow by taking limits along rational points that stay inside the common domain.

17.13 Exercise 17.13: Rational and irrational inputs with different values

What we must show. Prove (a) that the rational-indicator function is discontinuous everywhere, and (b) that the function equal to x at rational inputs and zero at irrational inputs is continuous exactly at zero.

Preparation: obtain both kinds of inputs near any real point.

Rationals are dense in $\mathbb{R}$. Irrationals are dense as well: given $u < v$, rational density gives a rational r in $(u - \sqrt{2}, v - \sqrt{2})$. Then $r + \sqrt{2} \in (u, v)$ is irrational, since rationality of that sum would make $\sqrt{2}$ the difference of two rationals.

For each real a and each n, choose a rational r_n and an irrational t_n in $(a - 1/n, a + 1/n)$. Both sequences tend to a, since their distances from a are less than $1/n$.

(a) The indicator takes incompatible limits along these inputs.

Let $f(x) = 1$ for rational x and $f(x) = 0$ for irrational x. At the two sequences just chosen, $f(r_n) = 1$ and $f(t_n) = 0$ for every n. If f were continuous at a, both image sequences would have to tend to $f(a)$. Their limits are 1 and 0, which are different. Thus f cannot be continuous at a. As a was arbitrary, it is discontinuous everywhere.

(b) At zero, the magnitude of the function is small with the input.

Let $h(x) = x$ for rational x and $h(x) = 0$ for irrational x. Then $h(0) = 0$, and in both cases $|h(x)| \leq |x|$. Given $\varepsilon > 0$, choose $\delta = \varepsilon$. For $|x| < \delta$,

$$|h(x) - h(0)| \leq |x| < \varepsilon.$$

Hence h is continuous at zero.

(b) At a nonzero point, the two dense input sets still disagree.

Fix $a \neq 0$ and choose the rational and irrational sequences from the preparation step. Then

$$h(r_n) = r_n \longrightarrow a, \qquad h(t_n) = 0 \longrightarrow 0.$$

Continuity at a would require both limits to equal $h(a)$. Because $a \neq 0$, the two limits differ, giving a contradiction.

Conclusion. The function in (a) is nowhere continuous. The function in (b) is continuous at zero and at no other point.

17.14 Exercise 17.14: A denominator-sensitive function on the rationals

What we must show. For nonzero rational $x = p/q$ in lowest terms with $q > 0$, let $f(x) = 1/q$; also let $f(0) = 1$ and let f vanish on irrationals. Prove continuity at every irrational point and discontinuity at every rational point.

Plan. Near an irrational point, only finitely many fractions in a bounded neighborhood have small denominators. Avoid those finitely many fractions; all remaining nonzero values of f are small.

Unify the notation at the rational point zero.

Write $0 = 0/1$ in lowest terms. The same denominator rule then gives $1/q = 1 = f(0)$. Thus all rational inputs, including zero, can be discussed with a positive reduced denominator.

Continuity at an irrational point: isolate the finitely many exceptions.

Fix $a \in \mathbb{R} \setminus \mathbb{Q}$ and $\varepsilon > 0$. Choose an integer $Q > \max\{1, 1/\varepsilon\}$, so $1/Q < \varepsilon$. Consider

$$F = \left\{ \frac{p}{q} \in [a - 1, a + 1] : p \in \mathbb{Z},\ 1 \leq q \leq Q,\ p/q \text{ is in lowest terms} \right\}.$$

This set is finite. Indeed, for a fraction in this set,

$$|p| = q\,|p/q| \leq q(|a| + 1) \leq Q(|a| + 1).$$

There are only finitely many possible integers p in this fixed bounded range, and only Q possible denominators. Their possible pairs, and hence the resulting fractions, are finite. The argument concerns a bounded neighborhood; fractions with small denominators are not a finite set on the whole real line.

No member of F equals a, because a is irrational. If $F \neq \varnothing$, let $d = \min\{|r - a| : r \in F\} > 0$ and set $\delta = \min\{1, d/2\}$. If $F = \varnothing$, set $\delta = 1$ instead. In either case $\delta > 0$.

Check every nearby input, including rational ones.

Suppose $|x - a| < \delta$. Then $x \in [a - 1, a + 1]$, and $x \notin F$ by the choice of δ. If x is irrational, $f(x) = 0 = f(a)$. If $x = p/q$ is rational in lowest terms, its denominator cannot be at most Q, since that would put x in F. Thus $q > Q$ and

$$|f(x) - f(a)| = \frac{1}{q} < \frac{1}{Q} < \varepsilon.$$

The same strict error bound is immediate in the irrational case. This proves continuity at a by Theorem 17.2.

Discontinuity at a rational point: approach through irrationals.

Fix a rational $a = p/q$ in lowest terms, allowing $0 = 0/1$. Then $f(a) = 1/q > 0$. By irrational density, choose irrational $t_n \in (a - 1/n, a + 1/n)$ for each n. These points tend to a, while $f(t_n) = 0$ for every n. The image limit is zero and differs from the positive value $f(a)$, so f is discontinuous at a.

Conclusion. The continuity set is exactly $\mathbb{R} \setminus \mathbb{Q}$; the discontinuity set is exactly $\mathbb{Q}$.

17.15 Exercise 17.15: A continuity test that omits the point itself

What we must show. Prove that f is continuous at a domain point x_0 if and only if $f(x_n) \to f(x_0)$ for every sequence in $\mathrm{dom}(f) \setminus \{x_0\}$ tending to x_0.

Forward implication: the omitted-point sequences are a subclass.

If f is continuous at x_0, Definition 17.1 gives the image limit for every domain sequence approaching x_0. It therefore gives it for each sequence that avoids x_0.

Reverse implication: split an arbitrary sequence by its equal terms.

Assume the omitted-point condition. Let (y_n) be any domain sequence with $y_n \to x_0$. At indices where $y_n = x_0$, the image error is exactly zero: $|f(y_n) - f(x_0)| = 0$. If only finitely many indices have $y_n \neq x_0$, the sequence is eventually equal to x_0, so its images are eventually equal to $f(x_0)$ and converge to that value.

If infinitely many indices have $y_n \neq x_0$, list all of them in increasing order as $n_1 < n_2 < \cdots$. The subsequence $x_k = y_{n_k}$ avoids x_0 and still tends to x_0. The assumption gives $f(x_k) \to f(x_0)$. Given $\varepsilon > 0$, choose an integer $K \geq 1$ so that $k > K$ implies $|f(x_k) - f(x_0)| < \varepsilon$. For any original index $n > n_K$, either $y_n = x_0$, giving zero error, or $n = n_k$ for some $k > K$, giving the same strict error bound by the selected subsequence. Thus $f(y_n) \to f(x_0)$ for the original full sequence. As the sequence was arbitrary, Definition 17.1 proves continuity.

Check what happens at an isolated domain point.

If some radius $r > 0$ satisfies

$$\mathrm{dom}(f) \cap (x_0 - r, x_0 + r) = \{x_0\},$$

there is no sequence of domain points avoiding x_0 that converges to x_0: convergence would eventually place its points in this ball, where no such points exist. The omitted-point condition is then vacuously true. Every domain sequence converging to x_0 is eventually x_0, so continuity is automatically true as well.

Conclusion. The punctured-sequence condition is equivalent to continuity, including at isolated points of the domain.

17.16 Exercise 17.16: Jumps in the postage-stamp function

What we must show. For fixed postage charges A and B, with the additional ounce charge $B > 0$, prove that the stated function P is discontinuous at every positive integer.

Plan. Approach an integer from the preceding interval. Its charge differs from the value at the integer by exactly B.

Write the domain and the two adjacent charges.

The domain is $[0, \infty)$. The charge is A on $[0, 1)$, and is $A + Bn$ on $[n, n + 1)$ for every positive integer n. Equivalently, the latter formula also covers $n = 0$. Fix any positive integer m. Its value is $P(m) = A + Bm$, because $m \in [m, m + 1)$. On the interval $[m - 1, m)$, the value is instead $A + B(m - 1)$.

Select domain points in the preceding interval.

For each $n \in \mathbb{N}$, set

$$x_n = m - \frac{1}{n+1}.$$

Since $0 < 1/(n + 1) < 1$, we have $m - 1 < x_n < m$, so x_n belongs to the domain and to the preceding charge interval. Also $|x_n - m| = 1/(n + 1) \to 0$, hence $x_n \to m$. But

$$P(x_n) = A + B(m - 1) \longrightarrow A + B(m - 1) \neq A + Bm = P(m).$$

The inequality holds because their difference is the nonzero charge B. Definition 17.1 therefore gives discontinuity at m. Since m was any positive integer, all such integers are discontinuity points.

Record the implicit nonzero-charge assumption.

The proof works for any fixed $B \neq 0$, although postage gives the natural assumption $B > 0$. If $B = 0$, the formula reduces to the constant function $P(x) = A$, which is continuous; the requested jump assertion necessarily presumes a nonzero additional-ounce charge. Charges here are fixed for the rate schedule under consideration.

Conclusion. With $B \neq 0$, each positive integer has a jump of magnitude $|B|$, so P is discontinuous at each of them.